

Data Retention & Decommissioning Policy

Sentinel Nexus — AI Data Lifecycle

RW

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Purpose

Per NIST AI 600-1 **GV-1.7-002** (decommissioning factors), **MS-2.10-001** (membership-inference / training-data extraction), and **MS-2.2-004** (privacy-enhancing techniques), this policy governs retention, expiry, and secure decommissioning of the AI-relevant data stores in Sentinel Nexus. It addresses control item **NC-4**.

Data stores in scope

Store	Contents	Sensitivity
Cold storage (Parquet on MinIO/S3)	Normalized sensor telemetry (Hive-partitioned)	High
Qdrant vector index	UEBA math vectors + payloads (per-investigation)	High
nexus_swarm_memory (Qdrant)	Verdict/immunity signatures + embeddings + incident reports	High
RSI / calibration / bias-audit / over-reliance ledgers	Cycle outcomes, calibration points, audit reports, accept-vs-override reliance points (NC-2/NC-8)	Moderate
Verdict-lineage ledger (NC-10)	Append-only SHA-256 hash chain over verdict/audit records	Moderate
Active-learning failure corpus (NC-9)	Misclassified / ungrounded verdicts captured as hard examples	Moderate
Energy-accounting ledger (NC-11)	Per-run energy (Wh) + carbon (gCO2e) estimates	Low
MLOps staging / training corpora	Synthetic + sanitized telemetry SFT records	Moderate

Retention & expiry

Store	Retention	Mechanism
Cold storage	Per data-classification schedule (operator-set)	S3 lifecycle / partition pruning
DR snapshots	7 days local	dr_snapshot timer

Store	Retention	Mechanism
nexus_swarm_memory immunity points	TTL-bounded (default 30 d, NEXUS_MEMORY_TTL_SECONDS)	A point past TTL no longer grants auto-dismissal (control implemented; recall delegates to memory_is_actionable)
Calibration / bias / over-reliance ledgers	Rolling; archived not deleted	Append-only files
Verdict-lineage ledger (NC-10)	Retained for the audit horizon; never edited (an edit breaks the hash chain by design)	Append-only, tamper-evident chain
Active-learning failure corpus (NC-9)	Retained until consumed into a training run, then archived	Append-only JSONL
Energy-accounting ledger (NC-11)	Rolling; aggregated into the metric plane	Append-only JSONL

Membership-inference posture (MS-2.10-001). nexus_swarm_memory stores embeddings of alert signatures. Risk: an adversary with read access could infer prior incidents. Mitigations: the store is inside the access-controlled enclave (no external exposure); the signature is keyed on stable *pattern identity* (sensor|source_type|vector), not raw PII; the TTL bounds the window. **Open action (POA&M-4):** add a periodic membership-inference review and evaluate differential-privacy / anonymization on the memory vectors.

Decommissioning (GV-1.7-002)

When a model, sensor source, or the system is retired, the following are verified:

1. **Data retention requirements** — confirm legal/operational holds before purge.
2. **Secure erasure** — cryptographic erase of weights/secrets; purge of the relevant nexus_swarm_memory signatures and cold-storage partitions; revoke Vault paths.
3. **Data leakage after decommission** — confirm no residual in DR snapshots beyond the retention window; rotate the integrity_secret and any shared keys.
4. **Dependencies** — check upstream/downstream IoT/AI/data dependencies before removal (e.g. a retired sensor's source_type routing and SIEM fanout mapping).
5. **Open-source / model artifacts** — remove from the model registry per the N=3 + referenced-version retention rule.
6. **Operator entanglement** — communicate deactivation, reasons, and alternative process to affected operators (per MG-2.4-001).

Privacy-enhancing handling

- Training corpora are **credential-scrubbed** at staging (01_spool_chatml, 01_spool_datasets) — passwords/keys/tokens masked before any record enters the

corpus.

- Outbound DLP scrubs RFC-1918 ranges and high-entropy secrets before any frontier egress; the cognitive sanitizer wraps adversary-controlled data as untrusted.
- TEVV / report retention follows the document-retention rule (GV-1.5-003): raw .log files are not committed; structured .md/.xml reports are retained for audit.

Review

Reviewed annually and on any change to a data store's contents, classification, or the external interconnection set.